



An Axiomatic Concurrency Model

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3

```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    } } }  
}
```



S



-

R  +

R



-



S



—



C



-

R  +

R  -



C



—





po

po







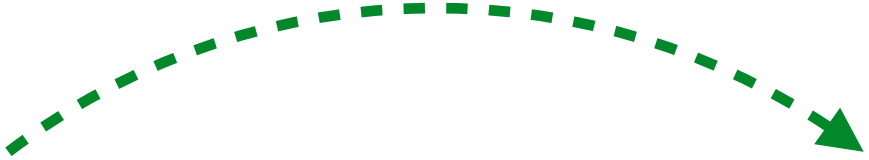
PO

po

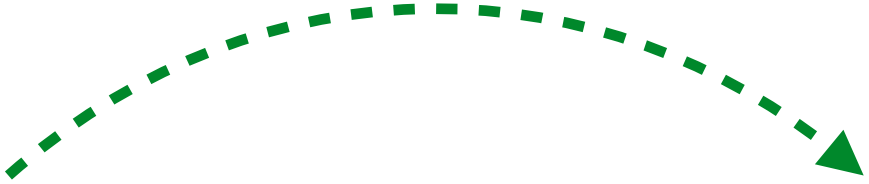


po

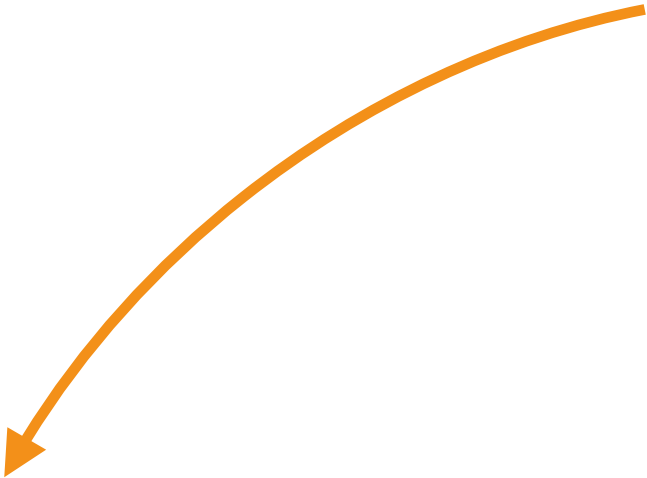
po



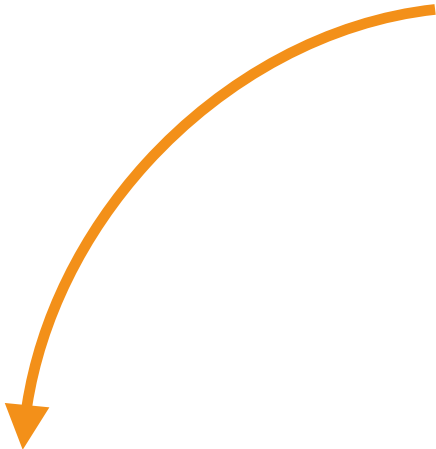








cosrc



collog

We derive the run relation from behaviour to run

```
update(src, +100)
```



```
update(src, -100)
```

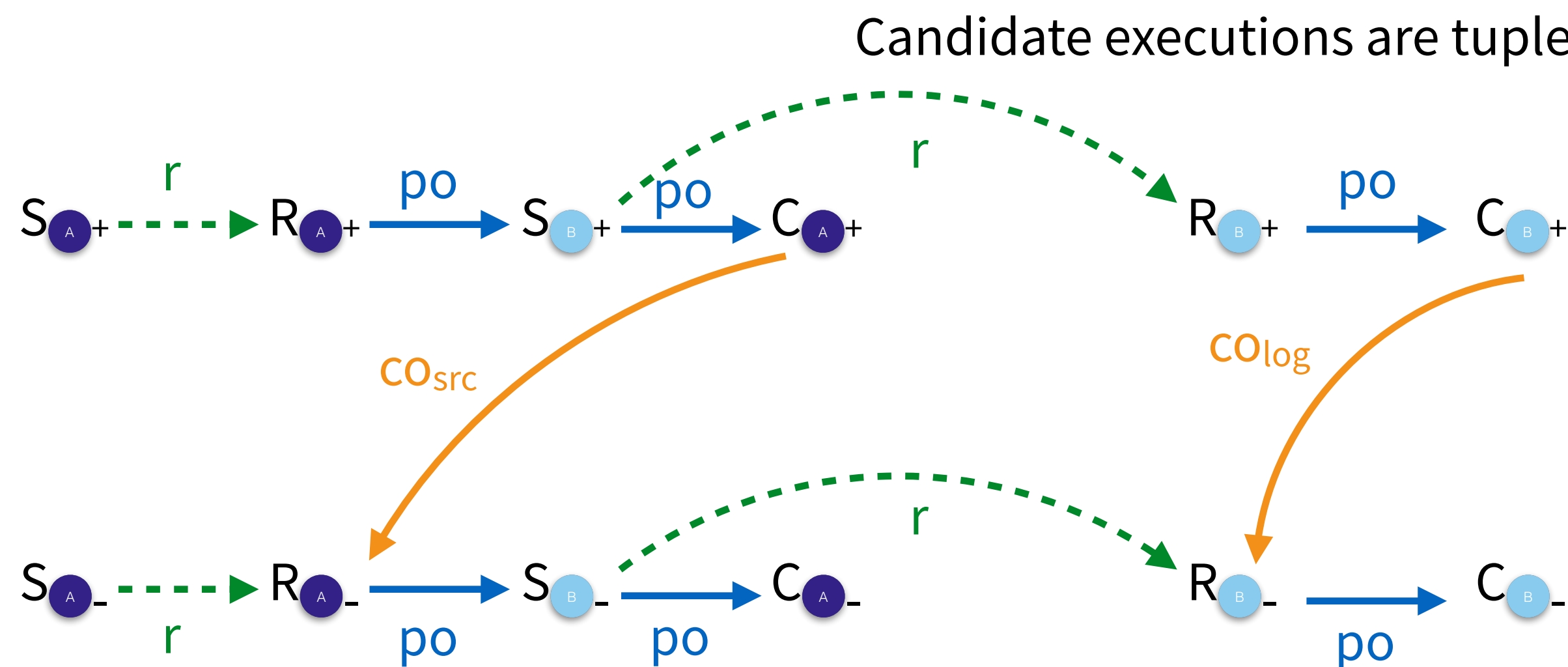
Candidate executions are tuples of (E, po, co)

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```
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      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    }  
  }  
}
```

update(src, +100)

update(src, -100)



We derive the run relation from behaviour spawn to run

An Axiomatic Concurrency Model

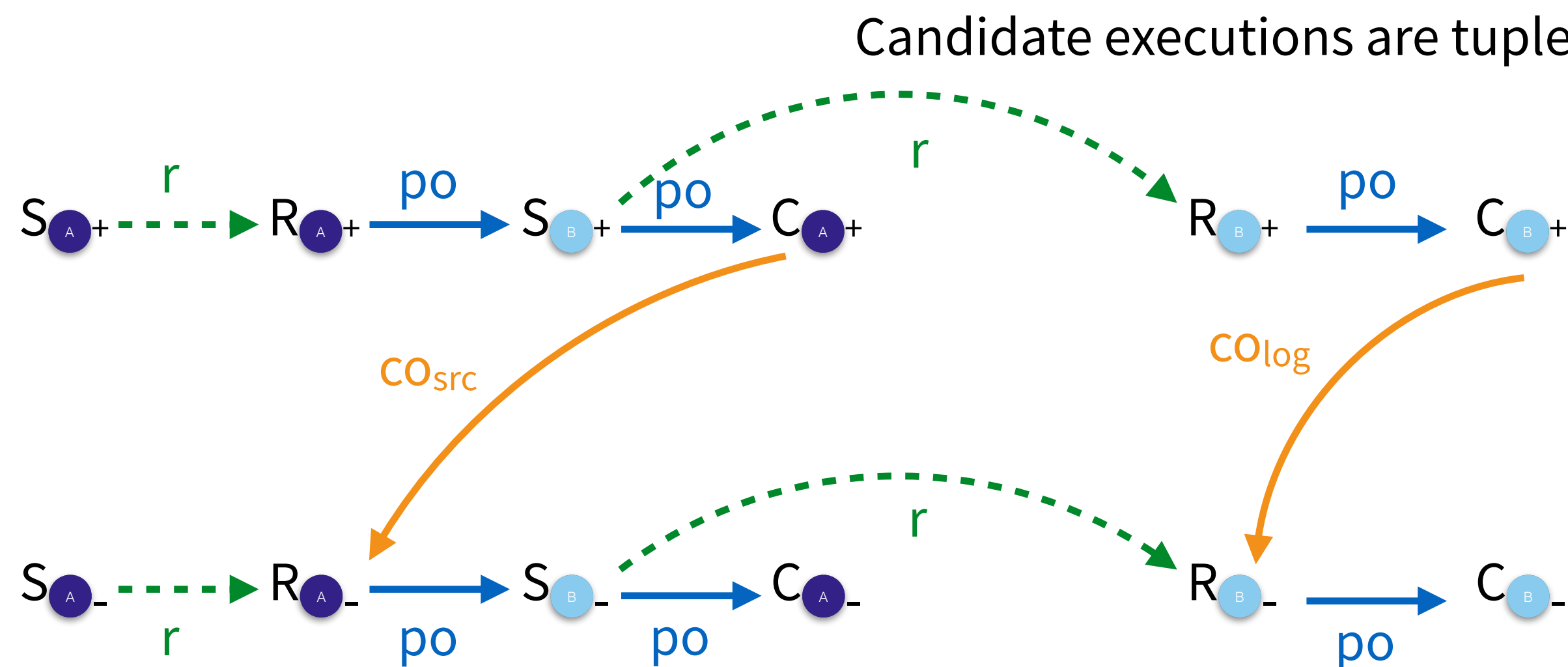
```

def update(account: cown[Account], amount: int) {
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    if(account.balance + amount > 0) {
      account.balance += amount
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      when(log) { B log.record(id, amount) }
    }
  }
}

```

update(src, +100)

update(src, -100)



We derive happens before

$$hb = \{(C_i, R_j) \mid S_i (po \cup r \cup co)^+ S_j \wedge \text{cowns}(i) \cap \text{cowns}(j) \neq \emptyset \dots\}$$

Is there a causal path between the spawns of two behaviours that require the same cowns?